

REVIEW OPEN ACCESS

The Genetics of Acne

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Received: 15 February 2025 | **Revised:** 7 July 2025 | **Accepted:** 11 July 2025

Funding: This work was supported by the National Medical Research Council CS-IRG (Clinician Scientist Individual Research Grant) MOH-000637.

Keywords: acne | genetics | GWAS | skin

ABSTRACT

This review addresses the genetics of acne vulgaris, the most common skin disease. It is characterized by the presence of comedones (blackheads), papules, and pustules. The condition is associated with sebaceous glands in the face and chest, which produce an oily substance called sebum. In developed nations, acne affects over 80% of adolescents. Mild disease usually resolves spontaneously. More severe acne can leave permanent, disfiguring scarring and strongly affects quality of life. In those cases, medical intervention is warranted. To date, antibiotics and retinoids (synthetic vitamin A derivatives) are the mainstays of treatment. Depending on the severity of the condition, these drugs may be administered either topically or systemically.

Whilst generally effective, they do come with significant drawbacks. Antibiotic use for treating acne is contributing to antimicrobial resistance. In addition, indiscriminate eradication of the skin microbiome negatively impacts skin health. Retinoids are teratogenic and have other undesirable side effects, such as skin irritation and increased UV sensitivity. Thus, there is a clear need for effective interventions that target the underlying disease mechanism, minimizing side effects.

Rapid progress has recently been made in understanding the mechanisms underlying acne. For decades, it was assumed that blackhead formation results from the accumulation of sebum in the hair follicle opening, due to increased sebum production at the onset of puberty. Subsequent colonization by the commensal bacterium *Cutibacterium acnes* then was thought to cause inflammation. It was also postulated that this micro-organism could induce blackheads. There are, however, several problems with this supposed sequence of events, not the least of which is that it doesn't explain how retinoids work, or why sebaceous glands associated with blackheads are atrophic and hence produce less sebum, not more.

Both GWAS and single gene disorders unequivocally indicate stem/progenitor cell maintenance and cellular migration as the most important processes in the pathogenesis of acne. Together with insights from mouse models, this new perspective is transforming the way we think about acne and its treatment.

1 | Introduction

This review discusses what is known about the genetics of acne, and what it teaches us about the pathogenesis of this common skin disorder (Zaenglein et al. 2016).

In developed nations, more than 80% of adolescents have acne, which is primarily characterized by the development of comedones (blackheads and whiteheads) (Zaenglein et al. 2016). Males and females are affected equally, although some work suggests that males have a slightly higher risk of developing the

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condition (Heng and Chew 2020). Most people have mild disease that resolves spontaneously. Around 20%–30% develop moderate to severe acne, and almost half of all patients will develop some degree of scarring (Liu et al. 2023b). Medical intervention is often required and mostly consists of antibiotics combined with synthetic vitamin A derivatives called retinoids (Zaenglein et al. 2016). This approach is generally effective; however, it has significant disadvantages. The use of antibiotics for acne measurably contributes to antimicrobial resistance (Dessinioti and Katsambas 2022). Retinoids are teratogenic and have other undesirable effects such as skin thinning and increased UV sensitivity (Bagatin and Costa 2020).

Acne typically affects adolescents but can persist well into adulthood with a reported prevalence of 12 % in females over the age of 25 (Holzmann and Shakery 2014). There are no published data for males, but a similar figure is presumed as males tend to less often seek medical assistance for skin concerns. As mentioned, acne is most prevalent in developed nations which has been attributed to the Western diet and lifestyle that come with increased affluence triggering the disease. Air pollution, in particular PM2.5, has also been implicated. There is some data to support this notion (Liu et al. 2018), for which evidence is otherwise rather scarce. Nonetheless, the geographic distribution clearly indicates that the interplay between genetics and environmental factors determines the disease risk (Saurat et al. 2024).

There is a clear need for effective and safe, non-antibiotic, and non-retinoid interventions. To develop those, the pathogenesis of acne needs to be understood at the cellular and molecular level. It is universally agreed that without androgens, there can be no acne (Kircik 2021). At the onset of puberty, the increase in plasma androgen levels causes sebaceous glands in the face and upper chest area to dramatically expand for reasons that remain unclear to this day. Accordingly, there is an increase in the production of sebum, an oily substance with roles in skin barrier maintenance, UV protection, and microbiome control (Dahlhoff et al. 2016). Until recently, blackheads were thought to be caused by the accumulation of sebum in the upper hair follicle. Subsequent colonization by the commensal bacterium *Cutibacterium acnes* (previously *Propionibacterium acnes*) was then postulated to cause inflammation (Zaenglein et al. 2016). Whilst superficially attractive, this proposed sequence of events has poor explanatory power. It does not explain the effectiveness of retinoids, nor does it offer a mechanism for the sebaceous gland atrophy which is associated with acne lesions. The contribution of *C. acnes* likewise is uncertain—for example, retinoids cure acne, yet they have no direct antibiotic action. People with acne have the same bacterial load as people without (Pécastaings et al. 2018). The model also does not explain why certain xenobiotics such as 2,3,7,8-Tetrachlorodibenzo-dioxin (TCDD, “dioxin”) can cause severe comedonic acne, in those cases referred to as “chloracne” (Saurat et al. 2012).

A paradigm shift came with the realization that the sebaceous gland grows from and is maintained by a population of progenitor cells situated in the junctional zone, where the gland empties into the hair canal. Ten years ago, Saurat proposed that these cells can assume either an infundibular keratinocyte or a sebocyte identity. In his model, blackheads result from sebaceous progenitors “switching” to the wrong identity: The “comedo

switch hypothesis” (Saurat 2015). This model is now understood to be essentially correct, with mostly genetic and, to a lesser extent, environmental factors triggering the switch (Clayton et al. 2019). Others and we have obtained good evidence that sebaceous progenitor identity is determined by Wnt signaling, with a stronger signal steering the cells toward an epidermal identity and predisposing to acne (Shang et al. 2022). As will be discussed later, GWAS does not suggest a prominent innate propensity to an inflammatory response in people with acne. However, recent work has indicated that in the course of disease, certain *C. acnes* phylotypes can outgrow others due to the altered milieu in the upper hair follicle (Dagnelie et al. 2019). This shift in phylotype distribution (dysbiosis) then causes the skin to mount a Th17-mediated inflammatory response which culminates in the appearance of pustules, papules, and nodules (Kistowska et al. 2015). The exact sequence of events in this response, the role of the skin microbiome and the reason why it does not affect everyone with acne remain poorly understood and are the subject of active research. This rather recent change of insight has come about because of fundamental work on single gene disorders that cause acne, combined with a better understanding of sebaceous gland biology (Common et al. 2019).

2 | Single Gene Disorders

A few rare monogenic disorders are associated with severe acne, starting at the onset of puberty. These conditions are of interest because they reveal molecular mechanisms underlying acne, which may be targeted for treatment. It is important in this context to note that the acne in all of these rare disorders responds well to standard treatment and is thus likely to have the same pathogenesis as its sporadic counterpart.

2.1 | Apert Syndrome (MIM #101200)

Apert syndrome is probably the best-known monogenic disorder associated with severe acne. There is strong, published evidence that the acne in Apert syndrome responds well to retinoids and oral contraceptives (Gilaberte et al. 2003; Hsieh and Ho 2005). The condition is caused by gain-of-function variants in the gene coding for fibroblast growth factor receptor 2 (FGFR2), a receptor tyrosine kinase. Almost every patient has one of two variants in exon 7, affecting an extracellular linker that connects Ig-like domains II and III (Matsumoto et al. 1998). Many other variants have been reported in *FGFR2* but only the ones associated with Apert syndrome will lead to severe acne. These variants abolish ligand specificity, resulting in aberrant and likely excessive receptor activation which may lead to the skin manifestations.

Further supporting the connection between acne and inappropriate FGFR2 signaling, there are reports of somatic mosaicism for *FGFR2* variants causing comedonic nevi following Blaschko lines (Munro and Wilkie 1998; Kiritsi et al. 2015). Despite these compelling clinical observations, the exact role of FGFR2 signaling in sebaceous gland development and maintenance remains unclear. There is much speculation, but little in the way of hard data although FGF signaling does show up in several GWAS (see there). It has been suggested that androgen stimulation of the

sebaceous gland at the onset of puberty is indirect—androgens induce skin fibroblasts to secrete FGFR2 ligands such as FGF1, which then initiate the proliferation of sebaceous progenitor cells (Melnik and Schmitz 2008). This scenario is consistent with our (unpublished) observation that primary human sebocytes kept in a progenitor state do not express the androgen receptor. However, there is no direct evidence that FGFR2 in humans is required for progenitor growth and maintenance. Observations in mice do indicate that *Fgfr2* is expressed in the sebaceous gland and is required for its normal development (Grose et al. 2007).

2.2 | Nevus Comedonicus (MIM #617025)

A tantalizing glimpse of the machinery underlying sebaceous differentiation is offered by the comedonic nevus caused by gain-of-function variants in *NEK9* (Levinsohn et al. 2016). NEK (NEver in mitosis Kinase) family members are involved in mitotic spindle positioning, affecting, among others, the direction of cell division (Bachus et al. 2022). NEK9 in particular has been implicated in mitotic spindle assembly (Casey et al. 2016). Its inappropriate activation in the skin disrupts normal follicular differentiation by causing a drastic, cystic expansion of the basal epidermal compartment at the expense of other structures (Levinsohn et al. 2016). Presumably, this is due to abnormal mitotic spindle orientation causing abnormally oriented cell divisions. The parallels with comedogenesis in acne are obvious and we propose that directional cell division is essential in the maintenance and differentiation of the sebaceous progenitor population. Any disruption may result in aberrant differentiation and, thence, cystic expansion manifesting as blackheads.

2.3 | Frank-Ter Haar and Winchester Syndromes (FTHS, WS; MIM #249420, #277950)

These similar but genetically distinct disorders both affect tissue remodeling. For this reason, we previously proposed grouping them in a single spectrum that we called DECORS, for “Defective Collagen Remodelling Spectrum” (de Vos et al. 2019). The key issue in both FTHS and WS is an inability to properly remodel connective tissue, including bone, to accommodate growth and environmental demands. Both conditions predispose to severe acne that responds to treatment with retinoids and antibiotics, just like sporadic disease (van Steensel et al. 2007).

FTHS is recessive, caused by loss of function variants in *SH3PXD2B*, which codes for the podosome adapter TKS4 (Wilson et al. 2014). Podosomes are actin-based protrusions that mobile cells such as macrophages generate when they need to move through a tissue, which requires digesting collagen and other extracellular matrix components (Chen et al. 2023). TKS4 recruits the membrane-bound matrix-metalloprotease MMP14 (Murphy and Courtneidge 2011), the loss of which causes WS (de Vos et al. 2018). It will come as no surprise that FTHS and WS have strongly overlapping symptoms (extensively reviewed in [de Vos et al. 2019]), which include the development of severe acne at the onset of puberty. The association is well established but has hardly been picked up in the dermatological literature. However, as we will

see it is a very important one because it reveals key aspects of sebaceous gland biology.

We also assigned MONA (multicentric osteolysis, nodulosis, and arthropathy, MIM #259600) to the DECORS spectrum (de Vos et al. 2019). It is caused by the loss of MMP2 (Martignetti et al. 2001), which is a direct target of MMP14 (de Vos et al. 2018). Hence, its symptoms closely align with those of FTHS and WS. However, acne has so far not been reported in MONA. This may well be due to ascertainment bias, as a connection with the rare genetic disorder is rarely considered when a patient presents with newly developing acne. This happened with the patients that we originally reported (van Steensel et al. 2007). They presented with cardiac symptoms. Their acne and other syndromic features were only noted later when the former became unusually severe.

The association between acne and DECORS indicates that tissue remodeling is an essential process in sebaceous gland development and homeostasis. The most parsimonious hypothesis consistent with our current understanding of the process is that sebaceous progenitor cells, upon directional cell division, move either into the gland (“downward”) or up into the infundibulum (“upward”). Their subsequent differentiation is determined by their location. Any failure of committed progenitor cells to migrate results in them staying in the junctional zone, where they will not receive the proper environmental clues. Consequently, they won’t differentiate into their eventual state and assume an “intermediate” fate, which is exactly what is seen in a comedo. Its lining is made up of cells that look like keratinocytes but contain significant amounts of lipids (Wilborn et al. 1978), which is unusual for a keratinocyte but expected for a sebocyte.

2.4 | Retinal Dystrophy, Iris Coloboma, and Comedogenic Acne Syndrome (RDCCAS; MIM #615147)

This extremely rare condition is caused by homozygous or heterozygous loss-of-function variants in the *RBP4* gene coding for retinol-binding protein 4 (Cukras et al. 2012). Loss of this lipocalin-family protein seems to impair the ability to deliver retinol from the liver to peripheral tissues. *RBP4* is also expressed in the retinal pigment epithelium, which might explain the eye symptoms. The skin may be particularly sensitive to the lack of *RBP4* because it requires high levels of retinol for its normal functioning (Carazo et al. 2021). It is not known why severe comedonic acne is such a prominent symptom, but it is consistent with the therapeutic efficacy of retinoids in treating acne. It has been hypothesized that this could at least in part be understood as supplementation (Zasada and Budzisz 2019). As will be discussed later, the central role of Wnt signaling in sebaceous gland development offers another clue—retinoids can antagonize Wnt signaling in the skin, for instance by inducing expression of DKK4 (Shang et al. 2022). Thus, it could be hypothesized that in RDCCAS, the (relative) absence of retinol in the skin leads to unopposed Wnt signaling and hence abnormal differentiation of sebaceous progenitors. Of note, this mechanism would also explain the other skin symptoms that have been reported in RDCCAS (and in vitamin A deficiency), such as

follicular hyperkeratosis with excessively dry skin (Carazo et al. 2021).

2.5 | Family, Twin, and Genetic Association Studies

Whilst monogenic disorders offer a peek into the pathobiology of acne, they have received little attention in the dermatological literature. Consequently, elucidation of their underlying gene defects has had little impact on the thinking about acne. Even so, it has long been recognized that there is a strong hereditary component to the disease and several studies have addressed this observation, albeit in a limited range of ethnicities. For example, a recent study (Greywal et al. 2022) enrolled 101 people with severe, scarring acne and documented their family history in detail. In this mostly Caucasian, male, and heavily pretreated cohort, it was found that 37.2% of subjects had one first-degree relative with moderate to severe scarring acne. A survey conducted at the world's largest gathering of twins (Annual Twins Day Festival in Twinsburg, Ohio) included 139 identical and fraternal twins as well as one triplet, for a total of 279 individuals. It was found that there were more concordant pairs in identical (64%) versus fraternal (49%) twins. These numbers are consistent with work carried out elsewhere (Mina-Vargas et al. 2017). Of note, most of these studies attribute 80% of disease variance to genetics, with the remaining 20% ascribed to environmental factors such as diet and air pollution.

Many other studies have addressed associations between one or a few genetic loci and the risk of developing acne. A recent systematic review identified 51 papers that identified 60 loci with a total of 100 variants implicated in acne (Heng and Chew 2020). Of note, most of these variants were in non-coding regions. The authors conducted a random-effect meta-analysis of all these previously identified risk loci and included their own Singapore Chinese population sample of 982 acne cases and 846 controls. It was found that most associations did not hold up, with inconsistent results between studies. For the ones that did, reduced acne risk was identified for people carrying certain alleles of *TNF* (rs1800629 A) and *CYP17A1* (rs743572 T), and an increased risk for the *FST* allele rs629725 A. The latter gene also gets picked up by GWAS (vide infra). The authors did identify 15 new SNPs in the 3'UTR of *TLR4* as a risk factor for developing acne.

Clearly, acne is a complex trait with a strong genetic component, a notion that is confirmed by genome-wide association studies (GWAS) that have been conducted over the past two decades.

2.6 | Genome-Wide Association Studies

To date, eight GWAS across different ethnicities have addressed the genetics of sporadic acne. These GWAS either used only severe acne cases for their analysis or found that the only clear genetic associations are the ones associated with severity. This is entirely consistent with the abovementioned observation that severe acne is more likely to be familial. It could also be taken to suggest that any environmental contribution is relatively minor, serving mostly as a trigger.

One of the first GWAS to be conducted for acne studied a relatively small Han Chinese cohort of 1056 cases with severe acne and 1056 controls (He et al. 2014). A replication study used an independent cohort of 1860 cases and 3660 controls. A strong point of this work was that all cases in the discovery stage of the GWAS were seen by dermatologists. Severe acne was defined as Pillsbury grade IV—meaning nodulocystic acne on the face and trunk. Two susceptibility loci were identified: *DDB2* (damage-specific DNA-binding protein 2) and *SELL* with roles in, respectively, androgen signaling and leucocyte migration. Thus, these are biologically plausible hits that, however, do not immediately suggest any novel insights into the pathogenesis of acne. The same group later verified *DDB2* by identity-by-descent analysis of their original GWAS data (Yang et al. 2019). They could not confirm *SELL* (although it was detected). They did identify *F13A1* and *ADH7* as potentially novel susceptibility loci, but neither gene has a plausible connection to acne. Another GWAS was conducted in 2014 in a relatively small ($n = 928$, of which 81 cases) European American population (Zhang et al. 2014). The diagnosis of severe acne was based on history, not direct examination. This study identified a variant in an SNP close to *MYC* (rs4133274) as a risk factor for severe teenage acne. This connection seems plausible, as animal models suggest a role for *Myc* in regulating androgen-dependent sebaceous gland growth (Cottle et al. 2013).

Another GWAS was published in 2014 and used 1893 UK cases with severe acne and 5132 controls (Navarini et al. 2014). The authors genotyped putative disease loci in a further 2063 acne cases and 1970 controls. This study identified three associations, all in genes connected to TGF β signaling: *OVOL1*, *FST*, and *TGFB2*. However intriguing, these findings did not immediately suggest novel insights into acne pathogenesis, although they do implicate tissue remodeling and scarring (Penn et al). All of these loci (but none of the ones identified in the Chinese studies) have been replicated in larger GWAS that were reported in 2018, beginning with Petridis et al. who performed a study of 3823 cases with severe acne recruited through hospital-based dermatologists in the UK, and 16,144 controls (Petridis et al. 2018). They followed it with a meta-analysis of a previous study with a sample size of 26,722. This study was the first to provide novel insights that aligned with Jean-Hilaire Saurat's prescient 2015 paper in which he proposed the comedo-switch model (Saurat 2015). The most prominent risk factor found by Petridis et al. was a gain of function allele (rs121908120 A) in *WNT10A*, which fits well with our observation in a mouse model that increasing Wnt signaling in the sebaceous progenitor compartment causes its cystic expansion with subsequent atrophy of the sebaceous gland (Shang et al. 2022). Further support for this role of Wnt signaling is provided by phenotype associated with loss of *WNT10A* in humans, odontonycho-dermal dysplasia (MIM #257980) which is associated with dry skin and sparse hair. More evidence for stem/progenitor cell involvement was provided by their finding of *LGR6* as a susceptibility locus. *LGR6* mediates WNT signaling (Barker and Clevers 2010). The identification of a risk allele in *FGF2* also fits with this narrative. Of note, *OVOL1*, which was first identified in the Navarini et al. (2014) study, and is a TGF β target, is also activated by WNT/ β -catenin (Li et al. 2002). *GLII* also shows up, as a well-known WNT target, as well as a mediator of Hedgehog signaling.

Further hits unrelated to WNT signaling include an intronic SNP in *SEMA4B* at a site that is known to bind TP63, a transcription factor that is essential for skin and appendage development (Mikkola 2007). *SEMA4B* itself codes for a semaphorin and belongs to a class of proteins involved, among other processes, in axonic guidance (Zang et al. 2021). This finding is rather intriguing, considering the importance of cell migration in the pathogenesis of acne. There is evidence that semaphorins can inhibit integrin-mediated attachment (Serini et al. 2003) and it is tempting to speculate that this role is relevant to acne. The identification of a risk allele in *LAMC2* supports this notion, as this gene codes for an adhesion molecule that can bind to integrins, in particular integrin $\alpha 6$ which is expressed by basal keratinocytes and sebocytes (Robinson et al. 2024).

Thus, from this GWAS, there emerges a picture of stem cell fate determination, cell migration, and tissue remodeling, all being relevant to the pathogenesis of acne, consistent with the story told by the monogenic disorders. It should be noted here that none of the loci identified in the study have any obvious connection with inflammation, which gives one pause.

Quite recently, large studies were published that both confirm the broad picture and add some interesting other angles. In 2022, the Simpson group published the results of a massive meta-analysis, performed in nine independent cohorts totaling 20,165 cases and 595,231 controls (Mitchell et al. 2022). The study identified 29 novel loci, in addition to confirming 14 out of 17 previously published ones. Several of the loci implicated cellular motility and adhesion, in keeping with previous results. For instance, genes implicated include *CSTA* coding for Cystatin A, which is a protease inhibitor known to be involved in intercellular adhesion (Blaydon et al. 2011). The acne risk allele is predicted to impair its function and, hence, cell migration in a manner reminiscent of the situation in the DECORS group of disorders. *EDAR* is a WNT signaling target that codes for the ectodysplasin receptor and is involved in skin appendage morphogenesis (Cui et al. 2014). The acne risk allele of this gene seems to be associated with curly hair in Japanese women (Endo et al. 2018).

Of note, this GWAS uncovered connections with inflammatory regulation by identifying a variant near *EDNRA*, and one near *IL36RN* eQTLs. The *EDNRA* gene codes for an endothelin receptor and missense variants in it cause mandibulofacial dysostosis with alopecia (MIM #616367). What is known about the *EDNRA* function is not connected to acne in any obvious way. However, the G protein *GNA3I* is a component of the endothelin signaling pathway and is known to interact with *SI1PR1* (Maeda et al. 2021), a G-protein coupled receptor involved in T-lymphocyte migration and survival. Hence, there might be a connection with T-cell responses which have been noted in acne as long as 37 years ago by Norris and Cunliffe, who observed early accumulation of CD4+ T-lymphocytes around comedones (Norris and Cunliffe 1988). This study identified another connection with inflammation in an acne susceptibility locus colocalized with *IL36RN* (interleukin 36 receptor antagonist) eQTLs. The risk allele is predicted to decrease *IL36RN* expression, in keeping with loss-of-function variants in *IL36RN* causing pustular psoriasis (Marrakchi et al. 2011). In this rare skin disease, pronounced skin infiltration with neutrophils occurs. This cell type predominates in acne pustules and can be targeted to reduce

inflammation, for instance with the antibiotic dapsone (diaminodiphenylsulfone) (Zhu and Stiller 2001), which is available in gel formulation.

The most recent GWAS was published in 2023, with largely confirmatory results and a few surprises. It was carried out in Estonia on 30,194 cases and 94,694 controls, followed by a meta-analysis of three independent European cohorts with a total of 34,422 affected individuals and 364,991 controls. This work identified 23 risk loci, 4 of which were novel. Of the remaining 23, 19 were firm replications and 7 were borderline significant. The 4 novel loci, combined with eQTL analysis, implicate the following genes: *FADS1*, *FADS2*, *LGR5*, *TSPAN8*, *FASN*, *CCDC57*, *SLC16A3*, *ZNR3*, *KREMEN*, and *CDC42BPG*.3.

LGR5, *ZNR3*, and *KREMEN1* are wingless signaling components, whereas *TSPAN8* codes for a member of the tetraspanin superfamily, a class of adhesion molecules. Interestingly, *TSPAN8* is a negative regulator of integrin β 1-mediated adhesion, just like *SEMA4B* (El Kharbili et al. 2017). *CDC42BPG* codes for a serine/threonine kinase that interacts with *CDC42*, a small GTPase which is a key player in cellular motility (Omble and Kulkarni 2022). *CCDC57* is a coiled-coil domain protein located at centrioles and microtubules, where it regulates cell division and, interestingly, ciliogenesis (Gurkaslar et al. 2020). Primary cilia are required, among others, for Hedgehog signaling and cell polarity (Bangs and Anderson 2017). There is no evidence yet that sebaceous progenitors have cilia. They might well do so, and we do have some unpublished data to support this notion. Collectively, these hits confirm the importance of cellular migration and adhesion in the pathobiology of acne.

The main novelty in this GWAS comes from the other genes that were identified. *FADS1*, *FADS2*, and *FASN* all code for enzymes involved in fatty acid metabolism, a crucial process in the production of sebum (Esler et al. 2019). Sebum composition in people with acne differs from that in people without, in having higher levels of monounsaturated fatty acids (MUFA) and altered ratios of triglycerides to wax esters (Okoro et al. 2021). There is no evidence, as yet, for a causal role for sebum composition in acne, and the observed changes might be due to other issues than enzyme dysfunction, such as altered microbiome composition (Dagnelie et al. 2019). However, others and we have shown that disrupting lipogenesis in the sebaceous progenitor compartment can lead to complete loss of the sebaceous gland (Lim et al. 2023). Hence, it might be speculated that deregulated lipid metabolism can contribute to abnormal sebaceous progenitor differentiation. Sebum production has been previously considered for acne treatment, however an *SCD1* inhibitor failed in clinical trials. Still, recent clinical trials show that *FASN* inhibition can improve acne and hence support the notion that there might be a connection with lipid metabolism (Xiang et al. 2024). A recent Mendelian randomization study used a GWAS of 486 serum metabolites from 7824 European adults to identify any that might be associated with acne (Wang et al. 2024). This work identified stearyl carnitine as a possible protective factor. Higher levels of stearyl carnitine are associated with a decreased number of CD4+ T-cells (Qian et al. 2021), the exact cell type that was long ago observed in early acne lesions (Jeremy et al. 2003). This is all correlation, and no mechanistic explanation was offered by the authors. Nonetheless, it is an intriguing observation that suggests further insights may

be gained from a closer look at the interaction between (lipid) metabolism and skin inflammation.

It should be noted here that the hits from GWAS conducted in China were never (firmly) replicated. It could be argued that ethnicity might explain this discrepancy. However, this seems unlikely. There is no evidence that the clinical presentation or response to treatment in Han Chinese is in any way different from people with other skin types. Also, the UK GWAS included multi-ethnic cohorts. An explanation might perhaps be found in ascertainment. There is no universally accepted system for acne severity grading. Hence, inhomogeneous distribution across severity grades might have masked some signals.

3 | Epigenetics

While this is a review of the genetics of acne, a few words on epigenetics may be in order. As opposed to genetic changes affecting acne risk, epigenetic modifications can be caused by environmental factors that can be modified and can be actioned with small molecule drugs. They are therefore of considerable interest.

Not much has been published yet on the epigenetics of acne. Recently, Liu et al. reported on their results with the analysis of DNA methylation in skin. They used the Infinium Methylation Beadchip, integrated with gene expression profiles and single-cell RNASeq in skin samples from people with acne (Liu et al. 2023a). “Healthy” skin samples served as controls, but the authors do not state whether these are age- and site-matched, or how they were obtained. There is also no mention of acne severity in study participants. The results of this study therefore need to be interpreted with some caution. That having been said, there are some interesting observations in it. Differential methylation affected 770 genes, with 274 hypermethylated (silenced) and 469 hypomethylated (upregulated). The paper did not specify which parts of the gene (including promoter region and 3' UTR) were differentially methylated. Most of the differentially methylated genes were suggested by GO analysis to be involved in “T-cell activation” (biological process), in addition to “cytokine–cytokine receptor interaction” and “chemokine signaling pathway”. These findings are consistent with the known involvement of T-cell signaling in acne (Kistowska et al. 2015). Some of the hypomethylated genes (and consequently, expressed at higher levels), such as *CD86*, *TLR4*, and *ITGAM* (coding for CD11b) implicate macrophages in the inflammatory response. This observation aligns with recent work that suggested a pivotal role for macrophages in the pathogenesis of acne (Do et al. 2022). Acne has a tendency to recur, and while site-specific recurrence of skin inflammation is usually attributed to resident T-memory cells (Trm) (Clark 2015), it is tempting to speculate that epigenetic reprogramming, or priming, of T-lymphocyte responses, has a role as well. The Liu et al. data remain to be replicated.

Wang et al. (2023) also addressed epigenetics in acne, but they assessed DNA methylation (with the Infinium BeadChip) in peripheral blood mononuclear cells to conduct an epigenome-wide association analysis. Their work used a total of 88 samples (44 patients, 44 controls), adding 5 more patients and 4 healthy

controls for differential expression analysis (Wang et al.). They identified 23 differentially methylated sites associated with severe acne; 16 were in genes and 7 were intergenic. Of particular note among the 16 genes are *IL1R1*, *MAPKAP2*, and *WNT9A/WNT14*. *IL1R1* (interleukin-1 receptor) is a receptor for interleukin-1 α and -1 β . Interleukin-1 α has been implicated in the pathogenesis of acne (Thanh et al. 2022). Guy and Kealy reported that exposing isolated, cultured pilosebaceous units (= hair follicles with the sebaceous gland attached) to this cytokine can cause infundibular hyperkeratosis, similar to what is found in comedones (Guy and Kealey 1998). Despite claims to the contrary (Thanh et al. 2022), interleukin-1 β probably has no role as there is no evidence that antagonists like Anakinra or Canakinumab ameliorate acne vulgaris. There is a neutralizing IL-1 α antibody in clinical trials, and it would be of interest to test its effect on acne. *MAPKAP2* is a serine/threonine kinase that is downstream of p38 MAPK, and is involved in the production of IL-1 α (Coulthard et al. 2009). It is intriguing that two genes from the same pathway show up in the epigenome analysis; it might be taken as (more) evidence for its involvement. *WNT9A* is mostly known for its role in endochondral ossification and joint development (Wang et al. 2014), and there are no immediately obvious connections to skin or skin disorders.

4 | Conclusion

GWAS, monogenic disorders, and animal models collectively provide overwhelming evidence in favor of the comedo switch hypothesis. Cellular adhesion and migration, plus stem or progenitor cell regulation take center stage, in sharp contrast to the traditional narrative that explains acne primarily as an inflammatory condition resulting from excess sebum production.

The genetics do not, unfortunately, explain the inflammation that so often occurs and which clearly needs to be seen as an epiphenomenon—something that occurs after comedones and sebaceous gland atrophy have set the stage. Epigenetics and (spatial) -omics techniques are more likely to give us the direly needed answers there. What has been published in this direction so far is interesting, but more rigorous, larger-scale studies are needed.

Meanwhile, with a clear handle on early pathogenesis, there is now an opportunity to use modern tools such as sebaceous glands and skin organoids to develop novel interventions. The ideal treatment is one that prevents acne from occurring in the first place, is cheap, easy to use, and preferably accessible as a personal care product. Thanks to all the advances made in acne genetics in just the past decade, this ideal may now be within reach.

Author Contributions

This manuscript was conceived and written entirely by the author.

Acknowledgments

The author acknowledges funding support for this work from the National Medical Research Council CS-IRG (Clinician Scientist Individual Research Grant) MOH-000637.

Conflicts of Interest

The author declares no conflicts of interest.

Data Availability Statement

Data sharing not applicable to this article as no datasets were generated or analysed during the current study.

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